Task-03: Hyperparameter Tuning, Cross Validation, and Comparison of Various Regression Models

In this task, I was asked to implement and evaluate the performances of **various regression models** on the **California Housing Dataset.** The main goal was to predict house prices based on the other features. Before implementing the model, various other steps were involved. These steps include:

- ☐ Importing the standard libraries and modules
- ☐ Importing the dataset
- ☐ Exploratory data analysis
- ☐ Standardization of the data/ Feature Scaling
- ☐ Splitting the data into test and train sets

For this study, three regression models were used: linear regression, ridge regression, and decision tree regressor. After performing all the data preprocessing steps, these models were trained on the standardized data and their predictions were recorded. Their performances were measured using the standard metrics such as RMSE and R2 Score.

METHODOLOGY

1. Exploratory Data Analysis:

Exploratory data analysis is the process of analysing, summarizing, and visualizing the dataset to understand the structure and spot anomalies before feeding the data to the machine learning models. In this task, I performed EDA in multiple steps:

1. Inspecting the dataset by checking the columns to know more about the dataset's features and its types and the range of values present.

2. Checking for missing values to either impute or drop such values

3. Calculating summarised statistics of the features to get a summary of the features

4. Visualising the distribution of features using histograms.

5. Correlation analysis to find the relationship between the features using correlation matrix and heatmap.

2. Feature Scaling and train-test split:

After completing EDA, the input data was standardised using standardscaler. Standardization or Z-score normalization involves rescaling of the features such that its standard deviation becomes 1 and mean becomes 0.

After feature scaling, the data was split into training and test sets using traintestsplit. It was split into an 80:20 ratio.

3. Model Training and Evaluation

The three models (linear regression, ridge regression, and decision tree) were trained on the training features and labels, and predictions were obtained on the basis of test features.
The following evaluation metrics were used:

- RMSE: root of the squared difference between the predicted and the actual values.
- R-Squared: represents the proportion of variance in the target variable that is explained by the model.

4. Overfitting Analysis

The decision tree model was trained on the training set and predictions were obtained for both training and test set. The mean squared error differed vastly for both these sets. Such difference indicates that the model was performing well on the training set but completely failed when applied to some unseen data. This means, the model overfit the training data.

4. Cross validation

A 5-fold cross validation was applied to the decision tree regressor. The dataset was divided into 5 equal sets. The decision tree model was trained for each fold. The 4 folds represent the labelled data, used to train the model, and the rest 1 fold contains unlabelled data to test the model. Every set acts as a test set at least once, i.e., this is an iterative process and the testing fold changes in every iteration. The final cross validation score is the average of the performance scores used for each fold(net mean squared error in this case).

5. Hyperparameter Tuning

To optimize the decision tree model's performance, hyperparameter tuning was performed using the GridSearchCV. The hyperparameters to be tuned were "max-depth" and "minimum-samples-split". In the param_grid, each hyperparameter was provided with a list of 5 potential values that it can assume. GridSearchCV performs cross validation on the decision tree model using these parameter values and saves the best performing parameters.

6. Saving the Model

After performance evaluation, a residual plot was created to compare the predictions of the best performing model to the actual values.
After that, the trained model was saved in a pickle file using joblib.

RESULTS

The following results were obtained after performance evaluation using RMSE and R2 Score:

	Models	RMSE	R2 Score
0	Linear Regression	0.745581	0.575788
1	Ridge Regression	0.702173	0.575791
2	Decision Tree Regression	0.702173	0.623791
3	Tuned Decision Tree	0.645363	0.682166

The tuned decision tree model turned out to be the best performing model.

DISCUSSION

Final Model Selection

The tuned decision tree model was selected as the best performing model because it performed well on both the training and test sets. It had a better RMSE and R2 score in comparison to the other models. The main problem with the decision tree model was overfitting. Hyperparameter tuning reduces this by reducing the complexity of the tree by fine tuning the hyperparameters such as max depth and minimum samples split. This, in turn, helped the model to generalize well on unseen data as unnecessary data patterns and noises were reduced.

Cross validation works by dividing the dataset into multiple folds and each fold is allowed to be used as a test set at least once. This makes cross validation results more reliable.

A simple model may generalise better, but it may not capture the underlying patterns that well, i.e., it is prone to the underfitting problem. Complex models, on the other hand, can understand the patterns in a lot better way, but may become prone to the overfitting problem. Hyperparameter tuning helps by balancing complexity and performance to achieve better generalisation results.

CONCLUSION

In this project, a standard machine learning pipeline was followed. This includes importing the dataset, exploratory data analysis, feature scaling, train-test split, model training, obtaining predictions, and evaluation of models on the basis of standard regression metrics.

After performing each test carefully, it is concluded that the fine-tuned decision tree was the best performer amongst the models, with an RMSE score of 0.645363 and R2 score of 0.682166.